

APPLICATIONS OF SIEVE METHODS IN FINITE FIELD PROJECTION THEORY

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ABSTRACT. We prove an analogue of the Linnik's large sieve in the finite field setting. We first prove a simple sieve theorem, which is able to give the same projection bound $|D| \lesssim \frac{Sq}{|X|}$ as the orthogonality/Fourier analysis approach. Afterwards, we try to obtain a "smarter" sieve that exploits the special projection properties of the set X to get new bounds.

1. INTRODUCTION

The main goal of this paper is to introduce analogues of Linnik's large sieve in finite-field settings. We first recall Linnik's theorem introduced in class:

Theorem 1.1 (Linnik's large sieve). *Let $f : \mathbb{Z} \rightarrow \mathbb{C}$ be a function supported in $[0, N]$. And let $P_M = \{p \in \mathbb{P} : \frac{M}{2} < p \leq M\}$. Then for any M*

$$\sum_{p \in P_M} \|(\pi_p f)_H\|_{L^2}^2 \lesssim \max\left(M, \frac{N}{M}\right) \|f_H\|_{L^2}^2$$

We want to obtain a similar sieve inequality. First, we analogously define what a projection of a function on finite fields means.

Definition 1.2. Let $\theta \in \mathbb{F}_q$ be a direction and let $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ be a function. Then $\pi_\theta f : \mathbb{F}_q \rightarrow \mathbb{C}$ is defined as

$$\pi_\theta f(x) = \sum_{\pi_\theta(y)=x} f(y)$$

for all $x \in \mathbb{F}_q$.

Recalling the definitions of high-frequency parts of functions on $\mathbb{F}_q$ and $\mathbb{F}_q^2$ from the lectures, we state our first result:

Theorem 1.3 (Simple sieve). *Let q be a prime power. Let $X \subset \mathbb{F}_q^2$ be a set of points, $D \subset \mathbb{F}_q$ be a set of directions, and say that $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. Then for any function $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ supported in X we have*

$$\sum_{\theta \in D} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim q \|f_H\|_{L^2}^2$$

The idea behind the proof of the simple sieve is to obtain a similar *Dictionary* that relates the Fourier transform of $\pi_\theta f$ to that of f . Then it turns out that in the Fourier sum we get, each non-zero frequency appears at most once. Hence, we get a straightforward bound on the Fourier sum.

As a corollary of Theorem 1.3, we obtain the following projection estimate by taking $f = \mathbf{1}_X$.

Corollary 1.4. *Let $X \subset \mathbb{F}_q^2$ be a set of points, $D \subset \mathbb{F}_q$ be a set of directions and say that $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. If $S \leq \frac{1}{2}q$ then $|D| \lesssim \frac{Sq}{|X|}$.*

The bound we get matches the bound we obtained in class using orthogonality/Fourier methods.

Our next inequality is what we call the smart sieve.

Theorem 1.5 (Smart Sieve). *Let q be a prime power, and let $X \subset \mathbb{F}_q^2$ be a set of points and $D \subset \mathbb{F}_q \cup \{\infty\}$ be a set of directions that contain 0 and ∞ . Write $D' = D \setminus \{0, \infty\}$. Let $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. Then for any function $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ supported in X we have*

$$\sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim (S|D| + S^{\frac{3}{2}}|D|^{\frac{1}{2}}) \|f\|_{L^2}^2$$

It turns out that by modifying the argument, we can slightly improve the above bound:

$$\sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim (|D| + S^{\frac{3}{2}}|D|^{\frac{1}{2}}) \|f\|_{L^2}^2$$

2. SIMPLE SIEVE

In this section we introduce our first result, which is the aforementioned simple sieve inequality.

Theorem 2.1 (Simple sieve). *Let q be a prime power. Let $X \subset \mathbb{F}_q^2$ be a set of points, $D \subset \mathbb{F}_q$ be a set of directions, and say that $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. Then for any function $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ supported in X we have*

$$\sum_{\theta \in D} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim q \|f_H\|_{L^2}^2$$

Before we begin the proof, recall that the proof of Nik's theorem relies on two crucial ideas:

- (i) *Dictionary:* From the definitions we get $\widehat{\pi_p f}(\alpha) = \widehat{f}(\frac{\alpha}{p})$ for any p prime and $\alpha \in \mathbb{Z}/p\mathbb{Z}$.
- (ii) *Locally constant:* Because $\text{supp } f \subseteq [0, N]$ it can be shown that $\widehat{f}(\xi)$ is locally constant on intervals of length $\sim \frac{1}{N}$.

After utilizing (i) we rewrite the quantity we want to bound to a sum of Fourier coefficients in $Q_M = \{\frac{\alpha}{p} : p \in P_M \text{ and } 0 < \alpha < p\}$. Then we relate this to the integral $\int_0^1 |\widehat{f}(\xi)|^2 d\xi$ using (ii).

We want to repeat the above procedure for our setting of finite fields. Firstly, the following Proposition gives *Dictionary* for our case.

Proposition 2.2. *For any $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ and $\theta \in \mathbb{F}_q$, we have*

$$\widehat{\pi_\theta f}(\psi) = \widehat{f}(\psi_\theta).$$

for all $\psi \in \mathbb{F}_q$. Here, $\psi_\theta \in \mathbb{F}_q^2$ is defined as $\psi_\theta = (\psi, \theta\psi)$.

Proof. As usual of the dictionary lemmas, we the proof follows from a straightforward definition checking:

$$\begin{aligned}\widehat{\pi_\theta f}(\psi) &= \sum_{x \in \mathbb{F}_q} \pi_\theta f(x) e(-x \cdot \psi) \\ &= \sum_{x \in \mathbb{F}_q} \left(\sum_{\pi_\theta(y)=x} f(y) \right) e(-x \cdot \psi) \\ &= \sum_{y \in \mathbb{F}_q^2} f(y) e(-\pi_\theta(y) \cdot \psi)\end{aligned}$$

Notice that $\pi_\theta(y) \cdot \psi = (y_1 + \theta y_2)\psi = y_1 \cdot \psi + y_2 \cdot \theta\psi = y \cdot \psi_\theta$. Hence we get

$$\widehat{\pi_\theta f}(\psi) = \sum_{y \in \mathbb{F}_q^2} f(y) e(-y \cdot \psi_\theta) = \widehat{f}(\psi_\theta)$$

□

It turns out that only the Dictionary is enough for the proof of Simple sieve (hence the name).

Proof of Theorem 2.1. Using the Dictionary Proposition 2.2, we rewrite

$$\begin{aligned}\sum_{\theta \in D} \|(\pi_\theta f)_H\|_{L^2}^2 &= \sum_{\theta \in D} \frac{1}{q} \sum_{\psi \neq 0} \left| \widehat{\pi_\theta f}(\psi) \right|^2 \\ &= \frac{1}{q} \sum_{\theta \in D} \sum_{\psi \neq 0} |\widehat{f}(\psi_\theta)|^2 \\ &= \frac{1}{q} \sum_{\xi \in Q_D} |\widehat{f}(\xi)|^2\end{aligned}$$

where Q_D is defined as $Q_D = \{(\psi, \theta\psi) : \theta \in D, \psi \in \mathbb{F}_q, \psi \neq 0\}$. Note that in the above, we are using the observation that every non-zero frequency ξ in $\mathbb{F}_q^2$ can be written as $(\psi, \theta\psi)$ for $\theta \in D$ and $\psi \in \mathbb{F}_q \setminus \{0\}$ at most once.

Since $Q_D \subseteq \mathbb{F}_q^2 \setminus \{0\}$, we get

$$\begin{aligned}\sum_{\theta \in D} \|(\pi_\theta f)_H\|_{L^2}^2 &= \frac{1}{q} \sum_{\xi \in Q_D} |\widehat{f}(\xi)|^2 \\ &\leq \frac{1}{q} \sum_{\xi \neq 0} |\widehat{f}(\xi)|^2 \\ &= q \|f_H\|_{L^2}^2\end{aligned}$$

as desired. □

2.1. Applications. We show a different proof of the following result presented in class:

Corollary 2.3. *Let $X \subset \mathbb{F}_q^2$ be a set of points, $D \subset \mathbb{F}_q$ be a set of directions and say that $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. If $S \leq \frac{1}{2}q$ then $|D| \lesssim \frac{Sq}{|X|}$.*

Proof. Take $f = \mathbf{1}_X$, which is clearly supported in X . The simple sieve states that

$$\sum_{\theta \in D} \|(\pi_\theta \mathbf{1}_X)_H\|_{L^2}^2 \lesssim q \|(\mathbf{1}_X)_H\|_{L^2}^2 \leq q|X|$$

On the other hand, $\pi_\theta \mathbf{1}_X$ is a function supported in a subset of size $\leq S$ in $\mathbb{F}_q$. We will prove the following lemma that we will use throughout this paper.

Lemma 2.4. *Let $g : \mathbb{F}_q \rightarrow \mathbb{C}$ be a non-negative function with $|\text{supp } g| \leq S$ and $\|g\|_{L^1} = M$. If $S \leq \frac{1}{2}q$ then $\|g_H\|_{L^2}^2 \gtrsim \frac{M^2}{S}$.*

Proof. Note that

$$g_0 = \frac{1}{q} \sum_{x \in \mathbb{F}_q} g(x) = \frac{M}{q}$$

so we see that

$$\|g_H\|_{L^2}^2 = \sum_{x \in \mathbb{F}_q} \left| g(x) - \frac{M}{q} \right|^2 \geq \sum_{x \in \text{supp } g} \left| g(x) - \frac{M}{q} \right|^2$$

Let $A = |\text{supp } g|$ be the size of the support of g . Then we have by Cauchy-Schwartz

$$\begin{aligned} \|g_H\|_{L^2}^2 &\geq \sum_{x \in \text{supp } g} \left| g(x) - \frac{M}{q} \right|^2 \\ &\geq \frac{1}{A} \left(\sum_{x \in \text{supp } g} \left| g(x) - \frac{M}{q} \right| \right)^2 \\ &\geq \frac{1}{A} \left(\sum_{x \in \text{supp } g} \left(g(x) - \frac{M}{q} \right) \right)^2 \\ &= \frac{1}{A} \left(M - M \frac{A}{q} \right)^2 \\ &= \frac{M^2}{A} \cdot \frac{(q-A)^2}{q^2} \end{aligned}$$

Remembering that $A \leq S \leq \frac{1}{2}q$, we get $\frac{(q-A)^2}{q^2} \sim 1$ and thus

$$\|g_H\|_{L^2}^2 \gtrsim \frac{M^2}{A} \geq \frac{M^2}{S}$$

as desired. $\square$

Now we return to the proof of Corollary 2.3. We apply the lemma to each of the functions $\pi_\theta \mathbf{1}_X$. Notice that

$$\|\pi_\theta \mathbf{1}_X\|_{L^1} = \sum_{x \in \mathbb{F}_q} \pi_\theta \mathbf{1}_X(x) = \sum_{y \in \mathbb{F}_q^2} \mathbf{1}_X(y) = |X|$$

Thus Lemma 2.4 gives $\|(\pi_\theta \mathbf{1}_X)_H\|_{L^2}^2 \gtrsim \frac{|X|^2}{S}$ and so we get

$$\sum_{\theta \in D} \|(\pi_\theta \mathbf{1}_X)_H\|_{L^2}^2 \gtrsim |D| \frac{|X|^2}{S}$$

Combined with our sieve bound, we get

$$|D| \frac{|X|^2}{S} \lesssim \sum_{\theta \in D} \|(\pi_\theta \mathbf{1}_X)_H\|_{L^2}^2 \lesssim q|X|$$

or $|D| \lesssim \frac{Sq}{|X|}$, as desired. $\square$

3. SMART SIEVE

In this section, we will first focus on showing the simpler smart sieve inequality:

Theorem 3.1. *Let q be a prime power, and let $X \subset \mathbb{F}_q^2$ be a set of points. Assume that $D \subset \mathbb{F}_q \cup \{\infty\}$ is a set of directions containing 0 and ∞ . Write $D' = D \setminus \{0, \infty\}$. Let $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. Then for any function $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ supported in X we have*

$$\sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim (S|D| + S^{\frac{3}{2}}|D|^{\frac{1}{2}}) \|f\|_{L^2}^2$$

Proof. We know that

$$\sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 = \frac{1}{q} \sum_{\xi \in Q_{D'}} |\widehat{f}(\xi)|^2$$

by the Dictionary lemmas.

Now, we construct our η function such that $f = f\eta$. Let η_0 and η_∞ be the following functions on $\mathbb{F}_q^2$:

$$\eta_0 = \sum_{y \in \pi_0(X)} \mathbf{1}_{L_{0,y}}$$

and

$$\eta_\infty = \sum_{y \in \pi_\infty(X)} \mathbf{1}_{L_{\infty,y}}$$

where we recall that for every $y \in \mathbb{F}_q$ and $\theta \in D$ we defined the line $L_{\theta,y}$ as

$$(3.1) \quad L_{\theta,y} = \{x \in \mathbb{F}_q^2 : \pi_\theta(x) = y\}$$

Then, we pick $\eta = \eta_0 \eta_\infty$. We can see that η is the characteristic function of the product set $\pi_0(X) \times \pi_\infty(X)$. One can compute that

$$\widehat{\mathbf{1}_{L_{0,y}}}(\xi) = \begin{cases} q \widehat{\mathbf{1}_{\pi_0(X)}}(\xi_1) & \text{if } \xi_2 = 0 \\ 0 & \text{otherwise} \end{cases}$$

and similarly that

$$\widehat{\mathbf{1}_{L_{\infty,y}}}(\xi) = \begin{cases} q \widehat{\mathbf{1}_{\pi_\infty(X)}}(\xi_2) & \text{if } \xi_1 = 0 \\ 0 & \text{otherwise} \end{cases}$$

From here, we take the Fourier Transform of $\eta = \eta_0 \eta_\infty$ and plug in above formulas to get

$$\begin{aligned} \widehat{\eta}(\xi) &= \frac{1}{q^2} \sum_{\omega \in \mathbb{F}_q^2} \widehat{\eta_0}(\omega) \widehat{\eta_\infty}(\xi - \omega) \\ &= \widehat{\mathbf{1}_{\pi_0(X)}}(\xi_1) \widehat{\mathbf{1}_{\pi_\infty(X)}}(\xi_2) \end{aligned}$$

Since it's clear that $\eta = 1$ on X (as $\eta_0 = \eta_\infty = 1$ on X) we know that $f = f\eta$. Hence we take the Fourier Transform to find that

$$\widehat{f}(\xi) = \frac{1}{q^2} \sum_{\omega \in \mathbb{F}_q^2} \widehat{f}(\omega) \widehat{\eta}(\xi - \omega)$$

We use the complex conjugate of this equation to get

$$|\widehat{f}(\xi)|^2 = \frac{1}{q^4} \sum_{\omega, \psi \in \mathbb{F}_q^2} \widehat{f}(\omega) \overline{\widehat{f}(\psi)} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)}$$

Using this expression, we write that

$$\begin{aligned} \sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 &= \frac{1}{q} \sum_{\xi \in Q_{D'}} |\widehat{f}(\xi)|^2 \\ &= \frac{1}{q^5} \sum_{\omega, \psi \in \mathbb{F}_q^2} \widehat{f}(\omega) \overline{\widehat{f}(\psi)} \left(\sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right) \end{aligned}$$

By triangle inequality and Cauchy-Schwartz, we get

$$\begin{aligned} \sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 &\leq \frac{1}{q^5} \sum_{\omega, \psi \in \mathbb{F}_q^2} |\widehat{f}(\omega)| |\widehat{f}(\psi)| \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right| \\ &\leq \frac{1}{q^5} \left(\sum_{\omega, \psi \in \mathbb{F}_q^2} |\widehat{f}(\omega)|^2 |\widehat{f}(\psi)|^2 \right)^{\frac{1}{2}} \\ &\quad \cdot \left(\sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 \right)^{\frac{1}{2}} \\ &= \frac{1}{q^5} \left(\sum_{\omega \in \mathbb{F}_q^2} |\widehat{f}(\omega)|^2 \right) \left(\sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 \right)^{\frac{1}{2}} \\ &= \frac{1}{q^3} \|f\|_{L^2}^2 \left(\sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 \right)^{\frac{1}{2}} \end{aligned}$$

Lastly, we evaluate the above expression and give an upper bound. It turns out that we can basically calculate it exactly:

$$\begin{aligned}
\sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 &= \sum_{\omega, \psi \in \mathbb{F}_q^2} \sum_{\xi, \chi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \cdot \overline{\widehat{\eta}(\chi - \omega)} \widehat{\eta}(\chi - \psi) \\
&= \sum_{\xi, \chi \in Q_{D'}} \left(\sum_{\omega \in \mathbb{F}_q^2} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\chi - \omega)} \right) \cdot \left(\sum_{\psi \in \mathbb{F}_q^2} \overline{\widehat{\eta}(\xi - \psi)} \widehat{\eta}(\chi - \psi) \right) \\
&= q^4 \sum_{\xi, \chi \in Q_{D'}} \left(\sum_{y \in \mathbb{F}_q^2} |\eta(y)|^2 e(-y \cdot (\xi - \chi)) \right) \cdot \left(\sum_{z \in \mathbb{F}_q^2} |\eta(z)|^2 e(z \cdot (\xi - \chi)) \right) \\
&= q^4 \sum_{y, z \in \mathbb{F}_q^2} |\eta(y)|^2 |\eta(z)|^2 \sum_{\xi, \chi \in Q_{D'}} e(-(y - z) \cdot (\xi - \chi)) \\
&= q^4 \sum_{y, z \in \mathbb{F}_q^2} |\eta(y)|^2 |\eta(z)|^2 \left| \sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) \right|^2 \\
&= q^4 \sum_{y, z \in T} \left| \sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) \right|^2
\end{aligned}$$

where we write $T = \pi_0(X) \times \pi_\infty(X) \subset \mathbb{F}_q^2$ such that $\eta = \mathbf{1}_T$.

Suppose that $\xi \in Q_{D'}$ has $\xi = (\alpha, \theta\alpha)$ for $\theta \in D'$ and $\alpha \neq 0$. Then we calculate that

$$\begin{aligned}
\xi \cdot (y - z) &= \alpha(y_1 - z_1) + \theta\alpha(y_2 - z_2) \\
&= \alpha(y_1 + \theta y_2) - \alpha(z_1 + \theta z_2) \\
&= \alpha \cdot (\pi_\theta(y) - \pi_\theta(z))
\end{aligned}$$

Using this, notice that

$$\begin{aligned}
\sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) &= \sum_{\theta \in D'} \sum_{\alpha \in \mathbb{F}_q, \alpha \neq 0} e(-\alpha \cdot (\pi_\theta(y) - \pi_\theta(z))) \\
&= \sum_{\theta \in D'} \begin{cases} q - 1 & \text{if } \pi_\theta(y) = \pi_\theta(z) \\ -1 & \text{otherwise} \end{cases}
\end{aligned}$$

If $y \neq z$ then there is at most one θ such that $\pi_\theta(y) = \pi_\theta(z)$. Write $y \sim_\theta z$ if $\pi_\theta(y) = \pi_\theta(z)$. Then we see that

$$\sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) = \begin{cases} (q-1)|D'| & \text{if } y = z \\ q - |D'| & \text{if } y \neq z \text{ and } y \sim_\theta z \text{ for some } \theta \in D' \\ -|D'| & \text{otherwise} \end{cases}$$

Next, we make the following Claim:

Lemma 3.2. *For every $\theta \in D'$, the number of pairs $y, z \in T$ with $y \neq z$ and $y \sim_\theta z$ is bounded above by $\lesssim S^3$.*

We will show this Claim later. For now, we use it to get

$$\begin{aligned} q^4 \sum_{y, z \in T} \left| \sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) \right|^2 &\lesssim q^4 (S^2 \cdot q^2 |D'|^2 + S^3 |D'| \cdot q^2 + S^4 \cdot |D'|^2) \\ &\lesssim q^6 (S^2 |D'|^2 + S^3 |D'|) \end{aligned}$$

Finally, plugging this in gets us

$$\begin{aligned} \sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 &= q^4 \sum_{y, z \in T} \left| \sum_{\xi \in Q_{D'}} e(-\xi \cdot (y - z)) \right|^2 \\ &\lesssim q^6 (S^2 |D'|^2 + S^3 |D'|) \end{aligned}$$

and thus

$$\begin{aligned} \sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 &\leq \frac{1}{q^3} \|f\|_{L^2}^2 \left(\sum_{\omega, \psi \in \mathbb{F}_q^2} \left| \sum_{\xi \in Q_{D'}} \widehat{\eta}(\xi - \omega) \overline{\widehat{\eta}(\xi - \psi)} \right|^2 \right)^{\frac{1}{2}} \\ &\lesssim \frac{1}{q^3} \|f\|_{L^2}^2 \cdot q^3 (S^2 |D'|^2 + S^3 |D'|)^{\frac{1}{2}} \\ &\lesssim (S |D'| + S^{\frac{3}{2}} |D'|^{\frac{1}{2}}) \|f\|_{L^2}^2 \end{aligned}$$

as desired. $\square$

Naturally, we wonder what this sieve inequality can give us for the projection theory problem. The following is what we get when we put $f = \mathbf{1}_X$:

Corollary 3.3. *Let $X \subset \mathbb{F}_q^2$ and $D \subset \mathbb{F}_q \cup \{\infty\}$. If $S = S(X, D) \leq \frac{1}{2} \min(q, |X|)$ and $|D| \geq 3$ then we have*

$$|D| \frac{|X|^2}{S} \lesssim |X| (S |D| + S^{\frac{3}{2}} |D|^{\frac{1}{2}})$$

and in particular

- (i) if $S \geq |D|$ then $|D| \lesssim \frac{S^5}{|X|^2}$.
- (ii) if $S \leq |D|$ then $|X| \lesssim S^2$.

Unfortunately, both of these bounds are obvious: clearly $|X| \leq S^2$ and in the case of $S \geq |D|$ we have $S^5 = S \cdot (S^2)^2 \geq |D| |X|^2$.

Turns out we can modify the proof slightly to get the following improved sieve inequality:

Theorem 3.4. *Let q be a prime power, and let $X \subset \mathbb{F}_q^2$ be a set of points. Assume that $D \subset \mathbb{F}_q \cup \{\infty\}$ be a set of directions such that $D = D' \cup \{0, \infty\}$. Let $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$. Then for any function $f : \mathbb{F}_q^2 \rightarrow \mathbb{C}$ supported in X we have*

$$\sum_{\theta \in D'} \|(\pi_\theta f)_H\|_{L^2}^2 \lesssim (|D| + S^{\frac{3}{2}} |D|^{\frac{1}{2}}) \|f\|_{L^2}^2$$

This slightly improved version gives the following corollaries:

Corollary 3.5. *Let $X \subset \mathbb{F}_q^2$ and $D \subset \mathbb{F}_q \cup \{\infty\}$. If $S = S(X, D) \leq \frac{1}{2} \min(q, |X|)$ and $|D| \geq 3$ we have*

$$|D| \frac{|X|^2}{S} \lesssim |X| (|D| + S^{\frac{3}{2}} |D|^{\frac{1}{2}})$$

which in particular gives

- (i) if $S \geq |D|^{\frac{1}{3}}$ then $|D| \lesssim \frac{S^5}{|X|^2}$
- (ii) if $S \leq |D|^{\frac{1}{3}}$ then $|X| \lesssim S$.

This implies the following theorem, which is sharp for the subfield example.

Theorem 3.6. *Let $X \subset \mathbb{F}_q^2$ and $D \subset \mathbb{F}_q \cup \{\infty\}$. Suppose that $S = S(X, D) \leq \frac{q}{2}$ and $|D| \geq 3$. There exists a constant $c > 0$ such that*

$$\text{if } S \leq c|X| \text{ then } |D| \lesssim \frac{S^5}{|X|^2}$$

4. CONNECTION TO INCIDENCE GEOMETRY

After heavy modification of my proof of the Smart Sieve, it seems like one can obtain the following combinatorial bound that depends on incidence counting:

Theorem 4.1. *Let q be a prime power and $X \subset \mathbb{F}_q^2$ be a set of points. Assume that $D \subset \mathbb{F}_q \cup \{\infty\}$ is a set of directions such that $D = D' \cup \{0, \infty\}$. We say that $S = S(X, D) = \max_{\theta \in D} |\pi_\theta(X)|$ and let $T = \pi_0(X) \times \pi_\infty(X)$ be a subset of $\mathbb{F}_q^2$. Also denote $P_{D'} = \{a \neq 0 : \pi_\theta(a) = 0 \text{ for some } \theta \in D'\}$.*